

With a strong foundation in software engineering and hands-on experience developing practical solutions across various domains, I enjoy taking on complex challenges through thoughtful design and implementation. Passionate about clean architecture, continuous learning, and creating technology that makes a meaningful impact. Enthusiast about music and audio development.

Experience

Linux Embedded Software Engineer, SolarEdge Technologies 2022-2025

- Contributed to SolarEdge's residential Smart Home Solution as a core R&D team member
- Developed backend services on a Linux-based, multi-device platform to ensure low latency and high performance
- Engineered energy-management algorithms balancing user demands with efficient utilization of multiple power sources
- Designed and maintained modular data structures and modules for seamless integration of diverse devices over various communication protocols
- Collaborated with software and hardware teams throughout all development stages to streamline debugging, maintenance, and deliver a flawless user experience
- Built and validated real-world hardware test environments to support rigorous QA and prototyping
- Primarily coded in C (with supplemental Python and C#), and implemented unit tests using CMocka

Official Basketball Referee, Israeli Basketball Association 2012-2014, 2017-2018

Education

Computer Science BSc., Technion 2019-2024

- Focus courses: Machine Learning, Deep Learning, Natural Language Processing and Program Synthesis
- Noticeable courses: Data structures, Compilation, Operating systems, Algorithms

Projects

Audio De-Reverberation tool (guided by Dr. Lior Arbel), [Dereverberation project](#)

Improved an audio enhancement system by integrating an LS-UNET architecture with a UNET model. Engineered a processing pipeline that significantly improved audio clarity by addressing both reverberation artifacts and background noise. Implemented custom deep learning components and audio processing tools from first principles, optimizing the system through strategic fine-tuning and hyperparameter optimization. The resulting solution demonstrated superior performance metrics compared to standard single-task models

BallPit- a quirky physics based MIDI generator, [BallPit](#)

A MIDI generator plugin that harnesses the chaos and unpredictability of balls colliding in an enclosed space. It provides many customization options and an elevated degree of user control, both in notation and rhythm.

Exodus delay- an advanced audio delay effect, [Exodus delay](#)

A real time "individual-instance" delay effect enabling the customization of each delay feedback line with varied effects and filtering, providing it with a unique character

Optimal Running Pace via Bio-feedback (guided by Mr. Eran Zituni, Vista lab)

A personalized music recommendation system using sports watch biofeedback, including real-time heart rate and cadence monitoring, to determine the optimal running pace for balancing exertion and endurance.

Military Service

Combat Medic, "Moran" Special Artillery Unit 2014-2017

Technical Skills

Development: C · C++ · Python · Assembly · JUCE · PyTorch · Librosa · CMocka

Tools & Infra: Linux · Windows · Git · Github Actions · CMake · MQTT · Wireshark · Jira · Figma

Languages: Hebrew - Native, English - Native, Japanese - Beginner